

Victoria Junior College
2018 H2 Biology Prelim Paper 4
Answer

Question 1

(a) (i) With reference to the indicator colour chart (on screen), record the colour of the contents of the three test tubes, as well as the pH of the contents, in a suitable table. [3]

- Complete table drawn with correct column headings – (from left to right) Temperature / °C, Colour, pH;;
- Correct matching of the colour observed with pH according to the colour chart shown;;
- Correct trend of pH (whole no. or 1 d.p., consistent) with tube B (100 °C) having the lowest pH (not lower than 6.5), tube A (50 °C) and C (room temperature) having similar pH (between 8 and 9);;

(ii) Using your biological knowledge, explain your results. [4]

- High kinetic energy at 100 °C → Denaturation of urease;;
- Hydrogen and ionic bonds disrupted;; (Rej. disulfide bond)
- Shape of active site not maintained and no longer complementary to urea;;
- Ref. to kinetic energy not high enough to cause denaturation at 50 °C → Able to form enzyme-substrate complex;;
- Ref. to different amount of ammonium carbonate produced resulting in different as ammonium carbonate is alkaline;;

(iii) Explain the ways in which you would have expected the result to be different if tubes A and B had been maintained at their respective temperatures for one hour instead of five minutes. [2]

- Correct expected result for both tubes;;
 - Tube A – Colour corresponding to a lower pH than that stated in (a)
 - Tube B – Remain the same as that stated in (a)
- Correct explanation;;
 - More kinetic energy as time of heating increases → More enzyme denaturation (in tube A)

(iv) State a variable that is not controlled in your procedure that is likely to affect the rate of enzyme reaction. [1]

- pH of the reaction mixture;;

(v) Other than controlling the variable you mentioned in (iv), using the same apparatus and materials, state three other ways in which you could improve the investigation into the effect of temperature on the activity of the enzyme urease. [3]

Any 3

- More temperatures for investigation;;

- Increase duration;;
- Fix room temperature;;
- Include repeats;;
- Boil at the end of 30 minutes to stop further enzymatic reaction;;

(b) (i) Plot a graph to show the effect of urea concentration on the rate of enzymatic reaction. [4]

- Independent variable on x-axis;;
- Both axes correctly labelled including units, y-axis = average ... ;;
- Axes scaled appropriately so that the graph takes up at least 50% of the grid and divisions are equidistant;;
- Average values from table correctly plotted as points joined by straight lines (dot-to-dot plot) or appropriate line of best fit as required by the data, with no extrapolation beyond extreme measured data;;

(ii) Comment on the effectiveness of the use of titrations with dilute sulfuric acid in the investigation as compared to the method used in (a). [1]

- Ref. to quantitative results increasing objectivity;;
- Ref. to ability to differentiate between small differences in the amount of ammonium carbonate formed;;

(iii) State the degree of freedom for the t-test.

- 4;;

Explain what is meant by:

(iv) significant difference.

- Idea that the difference is caused by an outside factor / not due to chance;;

(v) p value less than 0.05.

- Idea that there is less than 5% chance that the differences between the (observed) results occurred by chance;;

Or

- Idea that there is more than 95% chance that the differences between the (observed) results occurred due to an outside factor;;

Question 2

(a) Describe the distribution of the coloured contents within the cells. [1]

- Uniform distribution of coloured content within the cell;;

(b) (i) Record your measurements in 'graticule units' in an appropriate manner. [4]

Objective lens used:

1. Independent variable: cells in left most column;;
2. Dependent variable: Dimensions of A and B / graticule units in next column;;
3. Records reading consistently to nearest whole number, no decimal;;
4. For each cell, values of A > values of B,

- At 10x objective lens, 10-30 graticule units for A;;
- At 40x objective lens, 30-80 graticule units for A;;
- At 60x objective lens, 40-150 graticule units for A;;

(ii) Account fully for the change in appearance of the cells when placed in 1 mol dm⁻³ sodium nitrate solution. [4]

1. 1 mol dm⁻³ sodium nitrate solution has a more negative / lower water potential than the cell content;;
2. Water moves out from the cell (a region of higher water potential) into the sodium nitrate solution (a region of lower water potential) by osmosis;;
3. Volume of cytoplasm decrease / shrink as water leaves cells;;
4. loss of water caused the cell surface membrane to pull away from the cell wall / cells to undergo plasmolysis;;
5. retention of pigments in the vacuole because of its high molar mass / large size;;

(c) Describe the appearance of these cells compared with those in step 6. [1]

- The coloured contents / pigments have leaked out from many cells / for cells that contain the coloured contents, they appear to be more concentrated / more plasmolysed;;

(d) (i) Record your observations of the cells that were originally mounted in sodium nitrate solution. [2]

1. Volume of cytoplasm increases from the plasmolysed state;;
2. Cells recovered from plasmolysis to a large extent/ cytoplasm of cells are still slightly detached from the cell wall;;

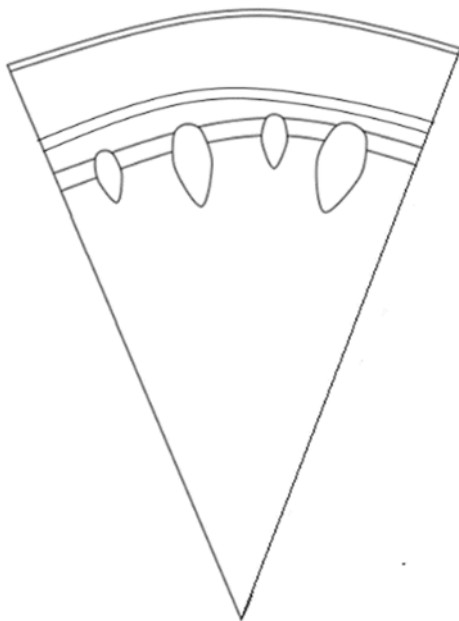
(ii) Record your observations of the cells that were originally mounted in lead nitrate solution, pointing out any differences that occurred between these cells and those that were originally mounted in sodium nitrate solution. [3]

1. Increase in cytoplasm length for cells in lead nitrate but no increase for cells in sodium nitrate;;
2. No recovery from plasmolysis;;
3. The protoplast was darker purple;;
4. The cells appeared granular;;
5. The protoplast has no shape / fragmented;;
6. Membrane appears to have burst/ more cells appeared colourless;;

(e) Explain your observations in (d)(i) and (ii) as fully as possible. [4]

1. Water has re-entered the sodium nitrate treated cell because membrane is still selectively permeable;
2. hence distilled water outside the cell which has a higher water potential compared to the cell content allows water to enter by osmosis;
3. whilst water did not move back into the lead nitrate treated cell due to loss of membrane selective permeability;
4. as lead ions disrupt structure of/ denature membrane protein / damage the membrane;

(f) Draw a large, plan diagram of a segment of the plant material in the space provided. [4]



1. Tissue section shown by thin, continuous line with no shading;
2. 3 to 5 vascular bundles, of correct shape and not uniform in size;
3. 3 layers: epidermis, parenchyma, with collenchyma (supporting tissue) separating parenchyma to 2 – 3 layers;
4. Correct proportion of above layers;
epidermis layer: at most same thickness as collenchyma,
first parenchyma layer ~ 10x epidermis,
VB: 3 – 10x of epidermis layer,

Question 3

(I) Theoretical consideration or rationale of the plan to justify the practical procedure;
[T1]

Enzyme lactase catalyses the decomposition of the substrate lactose to monosaccharides glucose and galactose.

The effectiveness of different methods of immobilisation of the lactase enzyme can be measured by packing equal sized burettes with the same amount of lactase (immobilised in different ways) and pouring a fixed volume of lactose into the burette. The amount of galactose product formed is determined by an enzymatic reaction using galactose oxidase that gives a 570nm product. The higher the absorbance / colour intensity at 570 nm, the more galactose product present and the more effective the method of lactase enzyme immobilisation.

(II) Procedure [P]

1. Independent variable – type of immobilisation of lactase
Dependent variable – absorbance/ colour intensity at 570 nm
2. Description of packing burettes with equal concentration lactase
3. Fix volume and concentration of lactose solution poured into burette
4. Drawing showing burette with immobilized lactase and collection of product, AND water bath for galactose oxidase reaction with galactose in filtrate collected from burette
5. Test concentration of galactose present using galactose oxidase. Fix time for reaction and temperature
6. Galactose oxidase to filtrate (galactose substrate) volume ratio 1:10
7. Use of colorimeter to determine amount of 570 nm product present
8. Control (mention how to set up, mention glass beads, alginate beads, cellulose)
9. Repeat the whole experiment twice more i.e. there should be minimum 3 readings;
10. Tabulation of table with headings (independent variable in leftmost column);
11. Graph with clear headings and labels (plot average absorbance against type of immobilisation);
12. State use of t-test carried out between different ways of immobilisation to determine if means are significantly different.
13. Risk assessment